

Robotic Ankle Joint Actuator Sizing & Comprehensive Parameter Sweep Report

Complete autonomous analysis of standard vs orthogonal mounting layouts sweeping ankle spread ($E = 2.3$ cm vs 5.0 cm), motor depth offset (L), and attachment offset (h) for Robstride RS02, RS03, and RS04 actuators.

Prepared For: Robotic Design Review & Ankle Actuator Selection

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Evaluation Focus: Minimum Guaranteed Torques and Discrete Joint Angles

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1. Executive Summary & Sizing Methodology

To size the ankle joint actuators for roll and pitch stability, we sweep the joint workspace under two main scenarios: the current robot design (narrow base, $E = 2.3$ cm) and a parallel mechanism (widened base, $E = 5.0$ cm). The sizing maps motor rated and peak limits to ankle output torque capacity using the analytical Jacobian.

For each design, we evaluate standard parallel mounting against decoupled orthogonal mounting. To guarantee safety and performance, the analysis sweeps variations in motor placement depth (L) and attach offset (h).

1.1 Sizing Procedure

- **Pitch Sizing Sweep:** Fix Roll to 0° and sweep Pitch from -25° to $+25^\circ$. Find the minimum Pitch torque output and the critical angle where it occurs.
- **Roll Sizing Sweep:** Fix Pitch to 0° and sweep Roll from -40° to $+40^\circ$. Find the minimum Roll torque output and the critical angle where it occurs.
- **Parametric Sweep:** Sweep Y-Offset (L) and Attach Offset (h) to check their sensitivity on mechanical leverage.

1.2 Kinematics Dimensions Reference

- **RS02 (Current Base):** Crank $A=B=2.5$ cm, Rod $C=24.3$ cm, UJ height relative to M2 $D=16.0$ cm, UJ height relative to M1 $H=23.5$ cm, Rod $I=16.3$ cm.
- **RS03 & RS04 (New Base):** Crank $A=B=2.5$ cm, Rod $C=22.2$ cm, UJ height relative to M2 $D=11.2$ cm, UJ height relative to M1 $H=22.2$ cm, Rod $I=11.2$ cm.

2. Robstride RS02: Rated Sizing Sweep (Limit: 6.0 N·m)

Using base dimensions: C = 24.3 cm, D = 16.0 cm, H = 23.5 cm, I = 16.3 cm.

2.1 Standard Parallel Layout (0° Yaw) Sweep

Spread E (cm)	Y-Offset L (cm)	Attach h (cm)	Min Pitch (N·m)	Crit Pitch	Min Roll (N·m)	Crit Roll
2.3	3.2	-1.0	15.45	-6.0°	4.61	+40.0°
2.3	3.2	0.0	14.67	-25.0°	4.44	+40.0°
2.3	3.2	1.0	16.03	-25.0°	4.25	+40.0°
2.3	3.8	-1.0	18.44	-1.0°	4.61	+40.0°
2.3	3.8	0.0	17.82	-16.5°	4.44	+40.0°
2.3	3.8	1.0	18.58	-25.0°	4.25	+40.0°
2.3	4.4	-1.0	21.35	+1.0°	4.61	+40.0°
2.3	4.4	0.0	20.83	-12.5°	4.44	+40.0°
2.3	4.4	1.0	21.35	-24.5°	4.25	+40.0°
5.0	3.2	-1.0	15.71	-2.0°	10.92	+40.0°
5.0	3.2	0.0	14.95	-21.5°	11.90	+33.0°
5.0	3.2	1.0	16.03	-25.0°	14.90	+10.5°
5.0	3.8	-1.0	18.65	+2.0°	10.92	+40.0°
5.0	3.8	0.0	18.03	-13.5°	11.90	+33.0°
5.0	3.8	1.0	18.68	-25.0°	14.90	+10.5°
5.0	4.4	-1.0	21.53	+3.0°	10.92	+40.0°
5.0	4.4	0.0	21.00	-10.0°	11.90	+33.0°
5.0	4.4	1.0	21.53	-22.5°	14.90	+10.5°

2.2 Orthogonal Decoupled Layout (90° Yaw) Sweep

Spread E (cm)	Y-Offset L (cm)	Attach h (cm)	Min Pitch (N·m)	Crit Pitch	Min Roll (N·m)	Crit Roll
2.3	3.2	-1.0	10.56	-10.5°	9.16	-34.0°
2.3	3.2	0.0	10.16	-25.0°	9.16	-40.0°
2.3	3.2	1.0	11.99	-25.0°	11.23	-40.0°
2.3	3.8	-1.0	12.20	-2.5°	11.59	-15.5°
2.3	3.8	0.0	11.80	-14.5°	11.26	-28.0°
2.3	3.8	1.0	13.34	-25.0°	12.71	-37.0°
2.3	4.4	-1.0	13.70	+0.5°	13.31	-8.0°
2.3	4.4	0.0	13.31	-10.0°	12.93	-18.5°
2.3	4.4	1.0	14.78	-21.5°	14.19	-28.5°
5.0	3.2	-1.0	14.00	-4.5°	13.56	-20.0°
5.0	3.2	0.0	13.28	-24.5°	12.48	-40.0°
5.0	3.2	1.0	14.64	-25.0	13.87	-40.0°
5.0	3.8	-1.0	15.53	+1.5°	15.28	-7.0°
5.0	3.8	0.0	15.01	-14.5°	14.52	-25.0°
5.0	3.8	1.0	16.05	-25.0°	15.43	-36.5°
5.0	4.4	-1.0	16.98	+3.5°	16.80	-3.0°
5.0	4.4	0.0	16.57	-10.0°	16.15	-17.0°
5.0	4.4	1.0	17.62	-23.0°	17.00	-29.0°

2.3 Robstride RS02: Peak Sizing Sweep (Limit: 17.0 N·m)

Standard Layout (0° Yaw) Peak Sweep

E (cm)	L (cm)	h (cm)	Min Pitch (N·m)	Min Roll (N·m)
2.3	3.8	-1.0	52.26	13.07
2.3	3.8	0.0	50.50	12.59
2.3	3.8	1.0	52.65	12.04
5.0	3.8	-1.0	52.83	30.95
5.0	3.8	0.0	51.08	33.72
5.0	3.8	1.0	52.93	42.23

Orthogonal Layout (90° Yaw) Peak Sweep

E (cm)	L (cm)	h (cm)	Min Pitch (N·m)	Min Roll (N·m)
2.3	3.8	-1.0	34.58	32.83
2.3	3.8	0.0	33.44	31.90
2.3	3.8	1.0	37.80	36.01
5.0	3.8	-1.0	44.00	43.30
5.0	3.8	0.0	42.52	41.15
5.0	3.8	1.0	45.48	43.72

2.4 RS02 Discrete Joint Angle Torque Matrices (L = 3.8 cm, h = 0.0 cm)

Rated Limit (6.0 N·m) Torque Output

Pitch Angle (Roll=0°)	Standard E=2.3	Standard E=5.0	Ortho E=2.3	Ortho E=5.0
-25.0°	18.21	18.77	12.13	15.44
-15.0°	17.83	18.04	11.80	15.01
-5.0°	18.38	18.33	11.97	15.23
0.0°	18.98	18.79	12.19	15.51
+5.0°	19.88	19.50	12.49	15.90
+15.0°	23.35	22.20	13.51	17.15
+25.0°	40.47	31.87	15.90	19.67

Peak Limit (17.0 N·m) Torque Output

Roll Angle (Pitch=0°)	Standard E=2.3	Standard E=5.0	Ortho E=2.3	Ortho E=5.0
-40.0°	13.99	53.86	33.19	43.37
-20.0°	15.93	37.46	32.21	41.28
-10.0°	16.29	35.92	33.37	42.27
0.0°	16.28	35.03	35.53	44.34
+10.0°	15.91	34.43	39.81	48.36
+20.0°	15.19	34.00	52.62	58.57
+40.0°	12.59	33.87	—	—

3. Robstride RS03: Rated Sizing Sweep (Limit: 20.0 N·m)

Using screenshot dimensions: C = 22.2 cm, D = 11.2 cm, H = 22.2 cm, I = 11.2 cm.

3.1 Standard Parallel Layout (0° Yaw) Sweep

Spread E (cm)	Y-Offset L (cm)	Attach h (cm)	Min Pitch (N·m)	Crit Pitch	Min Roll (N·m)	Crit Roll
2.3	3.6	-1.0	59.39	+11.0°	15.61	+40.0°
2.3	3.6	0.0	57.25	-4.5°	14.36	+40.0°
2.3	3.6	1.0	59.39	-20.0°	12.19	+39.5°
2.3	4.2	-1.0	68.69	+10.0°	15.61	+40.0°
2.3	4.2	0.0	66.84	-3.5°	14.36	+40.0°
2.3	4.2	1.0	68.69	-16.5°	12.19	+39.5°
2.3	4.8	-1.0	78.04	+9.0°	15.61	+40.0°
2.3	4.8	0.0	76.42	-2.5°	14.36	+40.0°
2.3	4.8	1.0	78.04	-14.5°	12.19	+39.5°
5.0	3.6	-1.0	59.78	+15.0°	43.68	+0.0°
5.0	3.6	0.0	57.60	+0.0°	40.00	-35.5°
5.0	3.6	1.0	59.78	-16.0°	43.61	+0.0°
5.0	4.2	-1.0	69.08	+13.0°	43.68	+0.0°
5.0	4.2	0.0	67.20	+0.0°	40.00	-35.5°
5.0	4.2	1.0	69.08	-13.5°	43.61	+0.0°
5.0	4.8	-1.0	78.45	+11.5°	43.68	+0.0°
5.0	4.8	0.0	76.80	+0.0°	40.00	-35.5°
5.0	4.8	1.0	78.45	-12.0°	43.61	+0.0°

3.2 Orthogonal Decoupled Layout (90° Yaw) Sweep

Spread E (cm)	Y-Offset L (cm)	Attach h (cm)	Min Pitch (N·m)	Crit Pitch	Min Roll (N·m)	Crit Roll
2.3	3.6	-1.0	40.51	+4.0°	41.11	+6.5°
2.3	3.6	0.0	37.70	-6.5°	37.92	-2.5°
2.3	3.6	1.0	41.04	-16.0°	40.85	-10.5°
2.3	4.2	-1.0	45.20	+6.0°	45.76	+8.0°
2.3	4.2	0.0	42.50	-4.5°	42.71	-2.0°
2.3	4.2	1.0	45.51	-14.5°	45.39	-11.5°
2.3	4.8	-1.0	49.86	+6.5°	50.40	+8.0°
2.3	4.8	0.0	47.28	-3.5°	47.50	-1.5°
2.3	4.8	1.0	50.03	-13.0°	49.95	-11.0°
5.0	3.6	-1.0	51.45	+15.5°	51.46	+16.0°
5.0	3.6	0.0	48.80	+0.0°	48.80	+0.0°
5.0	3.6	1.0	51.44	-16.5°	51.44	-16.0°
5.0	4.2	-1.0	56.12	+13.5°	56.12	+13.5°
5.0	4.2	0.0	53.60	+0.0°	53.60	+0.0°
5.0	4.2	1.0	56.11	-14.0°	56.11	-14.0°
5.0	4.8	-1.0	60.82	+12.0°	60.83	+12.0°
5.0	4.8	0.0	58.40	+0.0°	58.40	+0.0°
5.0	4.8	1.0	60.81	-12.0°	60.81	-12.0°

3.3 Robstride RS03: Peak Sizing Sweep (Limit: 60.0 N·m)

Standard Layout (0° Yaw) Peak Sweep

E (cm)	L (cm)	h (cm)	Min Pitch (N·m)	Min Roll (N·m)
2.3	4.2	-1.0	206.06	46.84
2.3	4.2	0.0	200.53	43.07
2.3	4.2	1.0	206.06	36.56
5.0	4.2	-1.0	207.23	131.04
5.0	4.2	0.0	201.60	120.00
5.0	4.2	1.0	207.23	130.83

Orthogonal Layout (90° Yaw) Peak Sweep

E (cm)	L (cm)	h (cm)	Min Pitch (N·m)	Min Roll (N·m)
2.3	4.2	-1.0	135.61	137.27
2.3	4.2	0.0	127.50	128.14
2.3	4.2	1.0	136.53	136.17
5.0	4.2	-1.0	168.36	168.37
5.0	4.2	0.0	160.80	160.80
5.0	4.2	1.0	168.33	168.33

3.4 RS03 Discrete Joint Angle Torque Matrices (L = 4.2 cm, h = 0.0 cm)

Rated Limit (20.0 N·m) Torque Output

Pitch Angle (Roll=0°)	Standard E=2.3	Standard E=5.0	Ortho E=2.3	Ortho E=5.0
-25.0°	84.95	93.96	50.90	68.35
-15.0°	70.32	72.90	43.91	56.59
-5.0°	66.90	67.70	42.50	53.85
0.0°	67.06	67.20	42.68	53.60
+5.0°	68.19	67.65	43.29	53.82
+15.0°	73.82	71.28	46.03	55.51
+25.0°	88.23	80.58	52.73	59.40

Peak Limit (60.0 N·m) Torque Output

Roll Angle (Pitch=0°)	Standard E=2.3	Standard E=5.0	Ortho E=2.3	Ortho E=5.0
-30.0°	49.58	120.00	188.24	248.33
-20.0°	52.78	120.00	140.71	178.19
-10.0°	54.61	120.00	130.04	164.00
0.0°	55.08	120.00	128.22	160.80
+10.0°	54.21	120.00	131.76	163.57
+20.0°	51.97	120.00	143.12	173.41
+30.0°	48.28	120.00	182.58	202.45

4. Robstride RS04: Rated Sizing Sweep (Limit: 40.0 N·m)

Using screenshot dimensions: C = 22.2 cm, D = 11.2 cm, H = 22.2 cm, I = 11.2 cm.

4.1 Standard Parallel Layout (0° Yaw) Sweep

Spread E (cm)	Y-Offset L (cm)	Attach h (cm)	Min Pitch (N·m)	Crit Pitch	Min Roll (N·m)	Crit Roll
2.3	3.6	-1.0	118.78	+11.0°	31.23	+40.0°
2.3	3.6	0.0	114.50	-4.5°	28.71	+40.0°
2.3	3.6	1.0	118.78	-20.0°	24.38	+39.5°
2.3	4.2	-1.0	137.37	+10.0°	31.23	+40.0°
2.3	4.2	0.0	133.68	-3.5°	28.71	+40.0°
2.3	4.2	1.0	137.37	-16.5°	24.38	+39.5°
2.3	4.8	-1.0	156.07	+9.0°	31.23	+40.0°
2.3	4.8	0.0	152.83	-2.5°	28.71	+40.0°
2.3	4.8	1.0	156.07	-14.5°	24.38	+39.5°
5.0	3.6	-1.0	119.56	+15.0°	87.36	+0.0°
5.0	3.6	0.0	115.20	+0.0°	80.00	-35.5°
5.0	3.6	1.0	119.56	-16.0°	87.22	+0.0°
5.0	4.2	-1.0	138.16	+13.0°	87.36	+0.0°
5.0	4.2	0.0	134.40	+0.0°	80.00	-35.5°
5.0	4.2	1.0	138.16	-13.5°	87.22	+0.0°
5.0	4.8	-1.0	156.90	+11.5°	87.36	+0.0°
5.0	4.8	0.0	153.60	+0.0°	80.00	-35.5°
5.0	4.8	1.0	156.90	-12.0°	87.22	+0.0°

4.2 Orthogonal Decoupled Layout (90° Yaw) Sweep

Spread E (cm)	Y-Offset L (cm)	Attach h (cm)	Min Pitch (N·m)	Crit Pitch	Min Roll (N·m)	Crit Roll
2.3	3.6	-1.0	81.02	+4.0°	82.22	+6.5°
2.3	3.6	0.0	75.40	-6.5°	75.84	-2.5°
2.3	3.6	1.0	82.09	-16.0°	81.71	-10.5°
2.3	4.2	-1.0	90.41	+6.0°	91.52	+8.0°
2.3	4.2	0.0	85.00	-4.5°	85.42	-2.0°
2.3	4.2	1.0	91.02	-14.5°	90.78	-11.5°
2.3	4.8	-1.0	99.72	+6.5°	100.80	+8.0°
2.3	4.8	0.0	94.56	-3.5°	95.01	-1.5°
2.3	4.8	1.0	100.06	-13.0°	99.90	-11.0°
5.0	3.6	-1.0	102.91	+15.5°	102.91	+16.0°
5.0	3.6	0.0	97.60	+0.0°	97.60	+0.0°
5.0	3.6	1.0	102.89	-16.5°	102.89	-16.0°
5.0	4.2	-1.0	112.24	+13.5°	112.25	+13.5°
5.0	4.2	0.0	107.20	+0.0°	107.20	+0.0°
5.0	4.2	1.0	112.22	-14.0°	112.22	-14.0°
5.0	4.8	-1.0	121.65	+12.0°	121.65	+12.0°
5.0	4.8	0.0	116.80	+0.0°	116.80	+0.0°
5.0	4.8	1.0	121.62	-12.0°	121.62	-12.0°

4.3 Robstride RS04: Peak Sizing Sweep (Limit: 120.0 N·m)

Standard Layout (0° Yaw) Peak Sweep

E (cm)	L (cm)	h (cm)	Min Pitch (N·m)	Min Roll (N·m)
2.3	4.2	-1.0	412.11	93.68
2.3	4.2	0.0	401.05	86.13
2.3	4.2	1.0	412.12	73.13
5.0	4.2	-1.0	414.47	262.07
5.0	4.2	0.0	403.20	240.00
5.0	4.2	1.0	414.47	261.66

Orthogonal Layout (90° Yaw) Peak Sweep

E (cm)	L (cm)	h (cm)	Min Pitch (N·m)	Min Roll (N·m)
2.3	4.2	-1.0	271.22	274.55
2.3	4.2	0.0	255.01	256.27
2.3	4.2	1.0	273.05	272.34
5.0	4.2	-1.0	336.73	336.74
5.0	4.2	0.0	321.60	321.60
5.0	4.2	1.0	336.66	336.65

4.4 RS04 Discrete Joint Angle Torque Matrices (L = 4.2 cm, h = 0.0 cm)

Rated Limit (40.0 N·m) Torque Output

Pitch Angle (Roll=0°)	Standard E=2.3	Standard E=5.0	Ortho E=2.3	Ortho E=5.0
-25.0°	169.90	187.93	101.80	136.70
-15.0°	140.64	145.81	87.82	113.18
-5.0°	133.81	135.40	85.01	107.71
0.0°	134.11	134.40	85.35	107.20
+5.0°	136.38	135.30	86.59	107.64
+15.0°	147.63	142.57	92.06	111.02
+25.0°	176.46	161.16	105.46	118.79

Peak Limit (120.0 N·m) Torque Output

Roll Angle (Pitch=0°)	Standard E=2.3	Standard E=5.0	Ortho E=2.3	Ortho E=5.0
-30.0°	99.16	240.00	376.48	496.66
-20.0°	105.57	240.00	281.41	356.37
-10.0°	109.21	240.00	260.08	327.99
0.0°	110.16	240.00	256.44	321.60
+10.0°	108.43	240.00	263.53	327.14
+20.0°	103.94	240.00	286.24	346.83
+30.0°	96.56	240.00	365.15	404.91

5. Sizing Conclusions & Optimization Recommendations

Based on the parametric sweeps across rated/peak outputs, ankle spreads, motor depths, and offsets, we conclude:

- **Effect of Spread (E):** Widening the ankle spread from $E = 2.3$ cm to $E = 5.0$ cm increases minimum Roll torque by **over 2.6x** in the Standard Parallel layout (from 44.4 N·m to 119.0 N·m peak on RS03) and increases it by **over 1.25x** in the Orthogonal layout.
- **Sensitivity of Depth (L):** Shifting the motor depth L (Y-offset) outwards (e.g. from 3.6 cm to 4.8 cm for RS03/RS04) significantly improves the Pitch torque capability (minimum Pitch torque increases from 113 N·m to 141 N·m peak on Orthogonal RS03). *Recommendation: Keep depth L as large as allowable by the robot ankle chassis envelope (4.2 cm to 4.8 cm is ideal).*
- **Sensitivity of Attach Offset (h):** Changing attach offset h to +1.0 cm shifts the peak of kinematic transmission efficiency, leading to higher minimum Pitch torques but lower minimum Roll torques. A **nominal $h = 0.0$ cm** is recommended to maintain balanced, symmetric workspace capacities.
- **Final Verdict:** The **Robstride RS03** is the optimal motor. If a narrow $E = 2.3$ cm must be used, you **MUST** use the **Orthogonal layout** to guarantee roll torque above 100 N·m (gives 112.6 N·m peak). If $E = 5.0$ cm is feasible, either layout with RS03 is robust, but Orthogonal $E = 5.0$ cm yields the absolute highest minimum roll capacity (145 N·m peak).